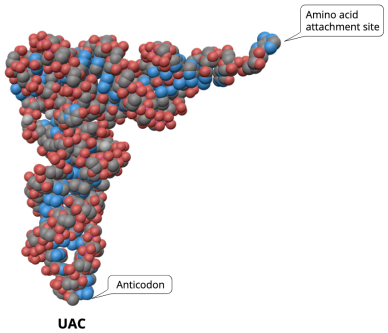
Translation

10/2/26

tRNAs

- The attachment site, which always has the nucleotide sequence CCA, is where amino acids bind so they can be carried by the tRNA
- The anticodon is a sequence of three bases that is complementary to a codon in an mRNA
- Different amino acids have different codons, so for each of the 20 amino acids, there is a different tRNA with a different anticodon
- Carry an amino acid to the correct codon in mRNA

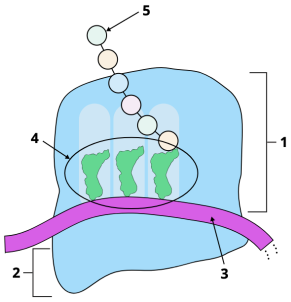


tRNAs consist of two subunits. The boundary between a large and small subunit is defined by where an mRNA strand is. When translation is not occurring, the subunits are separate. When it is occurring, the subunits bind together to form a single structure.

There is a channel in the machine where an mRNA enters one side, threads through the interior and exits out the other side. There are three sites in the middle of the machine where tRNAs fit. There is a channel in the machine, above the sites where tRNAs fit and where peptide bond formation occurs, where the newly synthesized protein emerges.

Translation Initiation

- Each mRNA contains the start codon AUG (codes for amino acid methionine)
- Requires a ribosome (specifically small subunit) and a tRNA with the anticodon UAC that is complementary to the AUG start codon in the mRNA and carries the amino acid methionine



Translation Elongation

- Once an mRNA has bound to the small subunit of a ribosome and a tRNA charged with methionine has bound to the start codon in the mRNA, the large subunit binds to the complex
- The sites inside ribosomes where tRNAs fit:
    - A - where amino acid charged tRNAs enter the ribosome. If the anticodon of a tRNA binds to the exposed codon in the mRNA, the tRNA and mRNA bind together

- P - the site where peptide bond formation takes place. The peptide bond forms between the amino acid held by the tRNA in the P site and the amino acid held by the tRNA in the A site
- E - where tRNAs sit once the amino acid they were carrying has been added to the growing protein. The now uncharged tRNA exits the ribosome from this site

Elongation is a three step sequence that keeps repeating

1. A tRNA enters the A site. If its anticodon is complementary to the next codon in the mRNA that needs to be translated, the tRNA binds to the mRNA and remains in the A-site.
2. The amino acids attached to the tRNA in the P site and to the tRNA in the A site are aligned in a way that places both amino acids in the machine's active site (this is where peptide bond formation takes place). After the peptide bond is formed, the linked amino acids are held by the tRNA in the A site, the growing protein is now one amino acid longer.
3. Proteins called elongation factors use energy to move the mRNA down three bases. This is important to the mRNA moving through the ribosome so the next codon in the sequence can be translated (this step is called translocation). After translocation, the now uncharged tRNA in the E site is ejected from the ribosome and the A site opens up. The empty A site allows a tRNA with the appropriate anticodon to bind to the next codon in the mRNA.